What the EEG-to-BOLD evidence supports

Resting EEG can support prediction research. Faithful regional reconstruction requires more than temporal correlation.

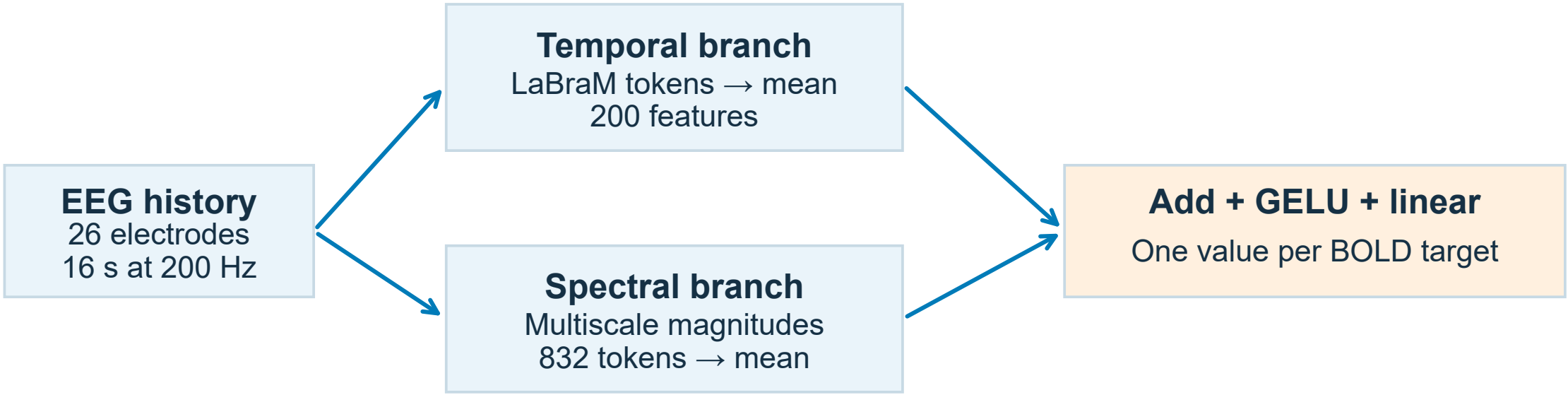
Existing ridge predictions: mean raw R^2 0.094 across 64 regions.

EEG predicts better than these physiology features.

The next architecture change is a controlled hypothesis, not a demonstrated improvement.

What NeuroBOLT computes

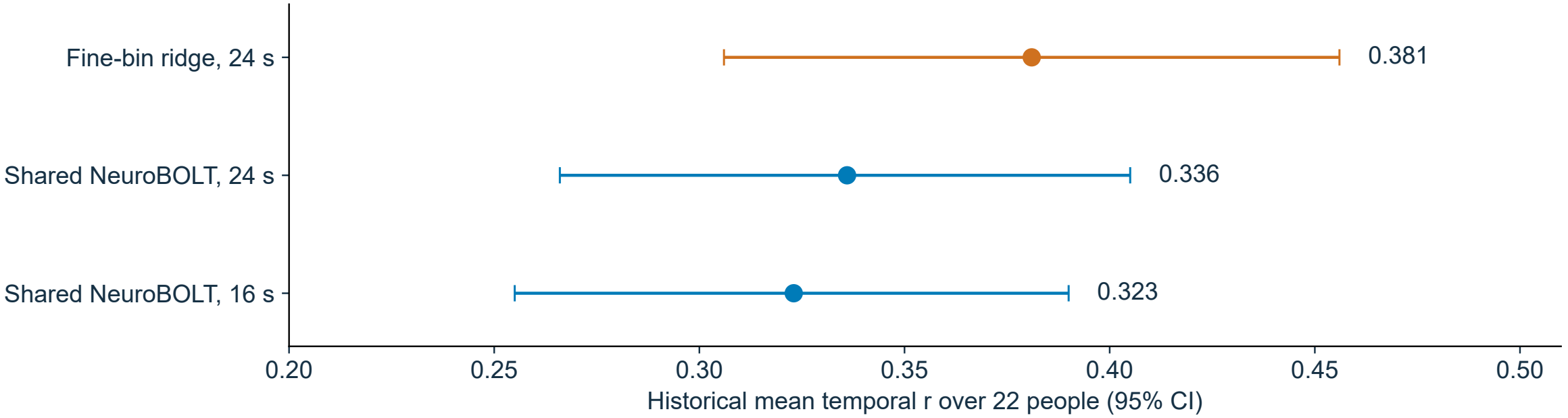
Waveform and spectral encoders summarize the same EEG history; the model adds their pooled representations.



At 24 seconds the spectral sequence has 1,248 tokens; dense time maps are learned indices.

The completed comparison still favours ridge

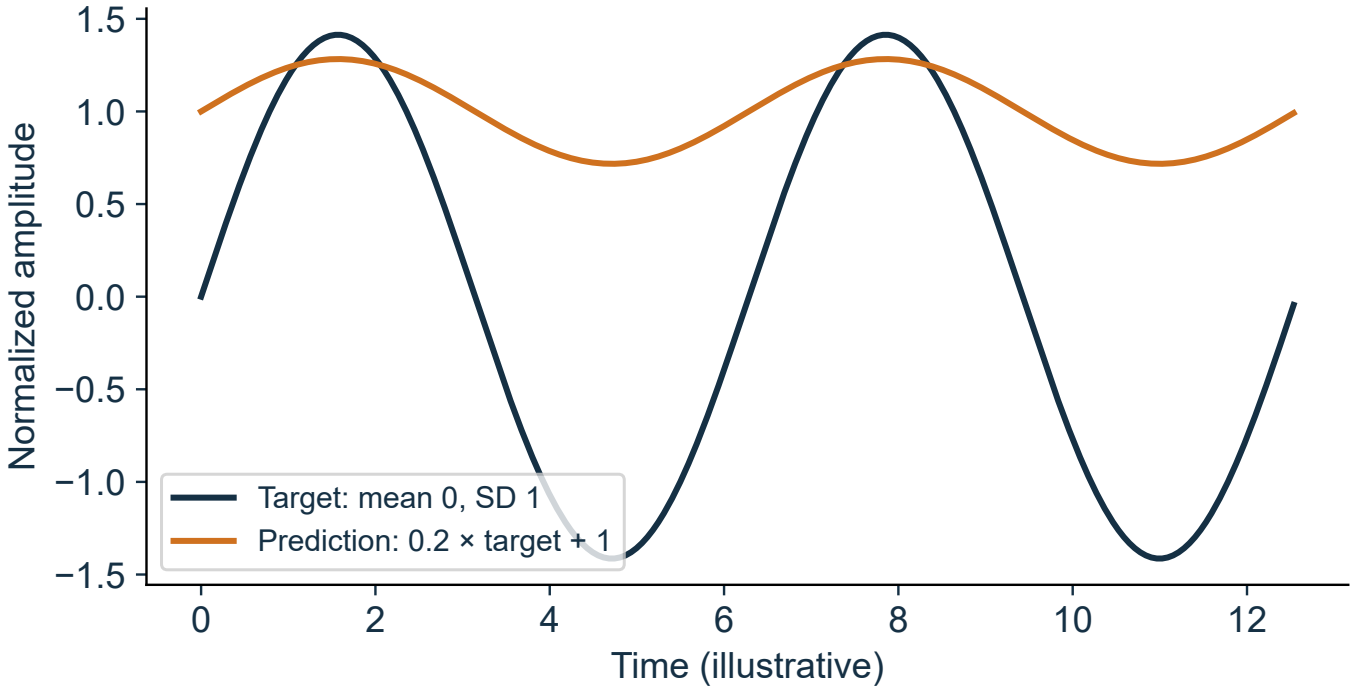
Historical five-fold temporal r: ridge .381; shared NeuroBOLT 24 s .336; shared NeuroBOLT 16 s .323.



Historical seven-target endpoint; distinct from the new 64-region raw-R^2 correction.

Correlation is not waveform accuracy

The earlier physiology partition used signed squared correlation under an R² label. The correction uses prediction error.



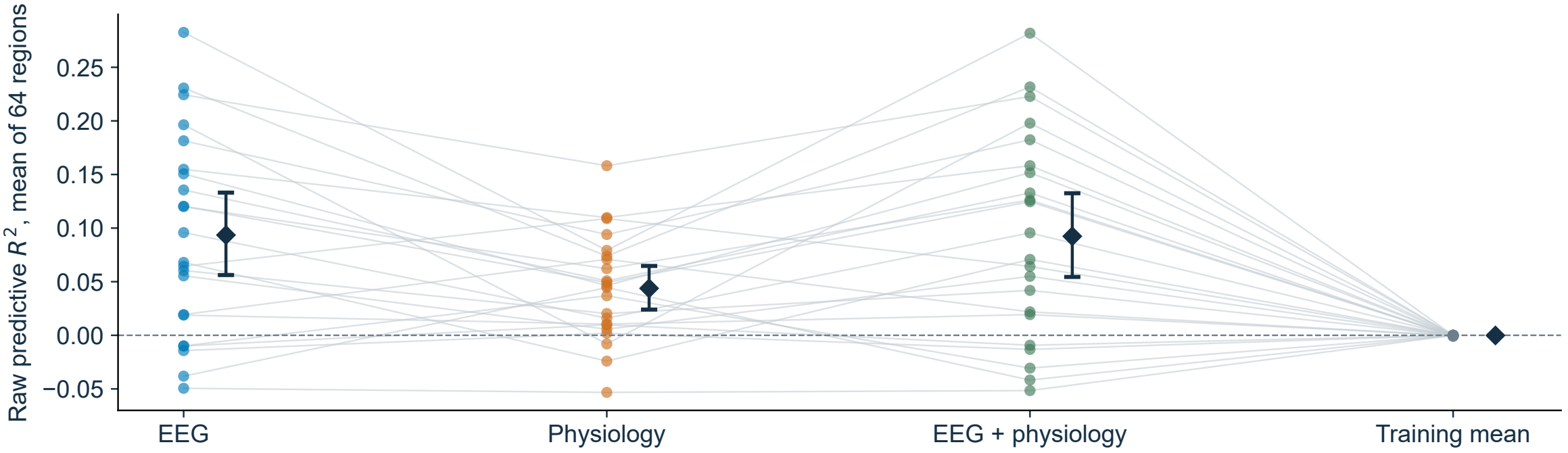
r = 1.00

Predictive $R^2 = -0.64$

Perfect cofluctuation can coexist with poor amplitude and offset.

People are the units of uncertainty

EEG: R² 0.094 ± 0.094; physiology 0.044 ± 0.049; joint 0.092 ± 0.096.

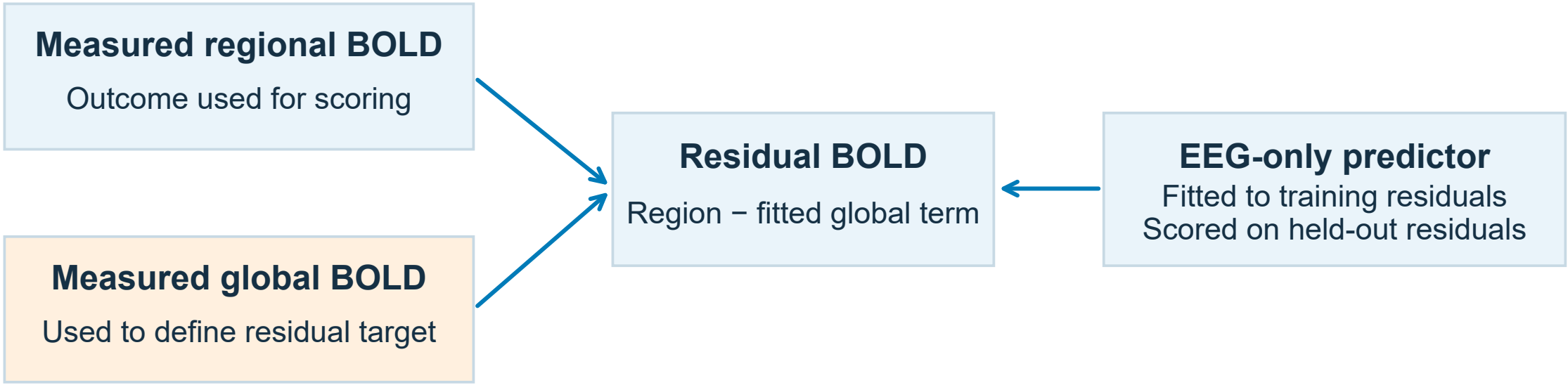


Dots: 22 people Diamonds: mean and participant-bootstrap 95% CI

EEG minus physiology: ΔR² +0.0497; 95% CI [0.0185, 0.0828]; p=0.0063; BY=0.0189; d_z=0.639; n=22.

Global-signal regression defines another target

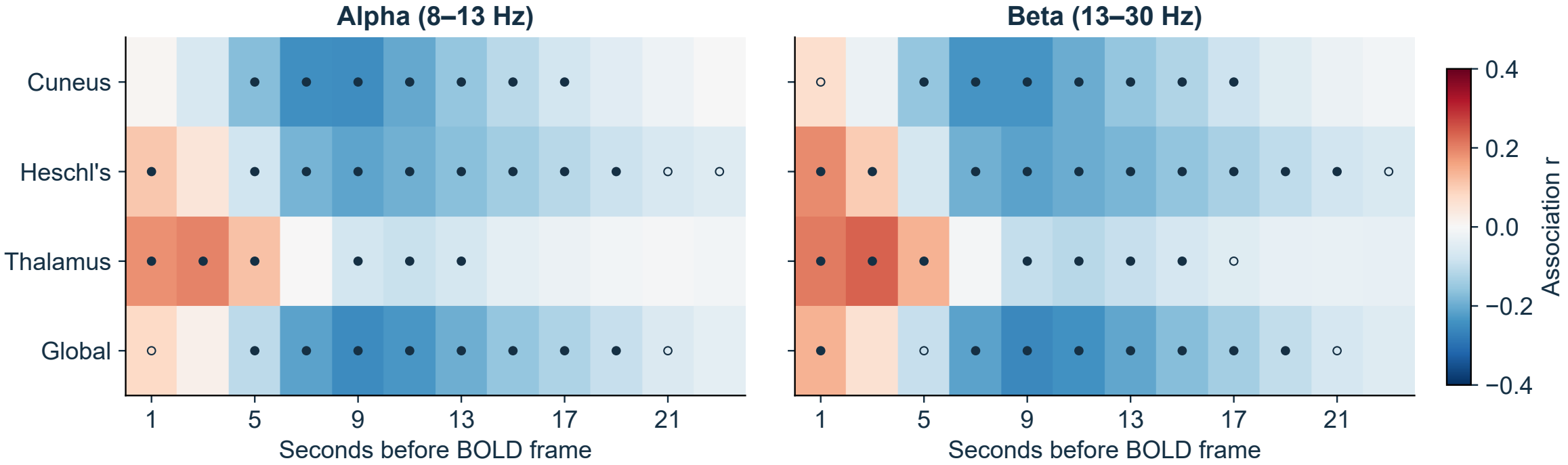
For existing residual-trained ridge, mean regional r=0.144, raw R^2=0.010. GSR is retrospective.



Measured test global BOLD is an outcome-definition input, never an EEG-model input.

Lag patterns need participant uncertainty

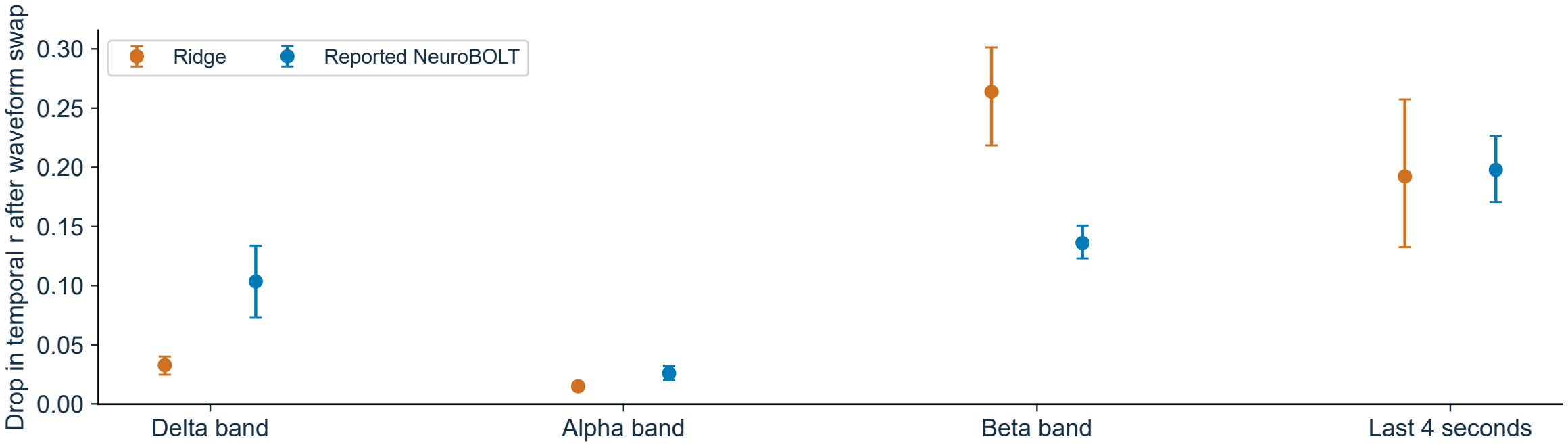
Alpha peak difference, thalamus minus cortex: -6 s; bootstrap interval [-8, -4] s. Resolution: 2 s.



22 people; 96 tests. Open circle: BH < .05; filled: BY < .05. Patterns are associations.

The checkpoint gate limits neural claims

CKA/IG were blocked by missing CV checkpoints. Existing waveform swaps below remain descriptive evidence from five people.



Historical swaps, 5 people. Published-checkpoint curve omitted: training provenance unresolved.

Next test: physical spectral-lag alignment

Replace one dense time-axis mapping in spectral-only NeuroBOLT24 with physical support-overlap weights.

Current spectral mapping

Dense learned mixtures of scale-window indices

→

Proposed controlled replacement

Fixed support-overlap mapping
48 half-second bins

Keep the remainder fixed

24 s spectral-only model, same frames, seeds, loss, encoder and readout

Removing 4,512 parameters is a regularization confound. This change remains untested.

A two-week investigation with a stopping rule

Complete one controlled change and explain its result. Expand only after a profiled comparison and a new compute budget.

